

Neel Shah

Quantitative Modeling · Forecasting & Calibration · ML Engineering

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github.com/neelshah/court-vision | linkedin.com/in/neelshah22

SUMMARY

I build calibrated pricing models and the tools that check them. My project, CourtVision, prices **four sports (NBA, MLB, soccer, tennis)** and scores every model **leak-free and walk-forward against the Shin-devigged closing line**. The honest result: on moneylines it lands near the close but never beats it, and on totals and tennis it trails. I also built the **validation harness** (walk-forward with purge and embargo, game-clustered Diebold-Mariano tests, a two-corpus acceptance rule) that **caught and retracted six of my own inflated results**, and I publish the rejects. Execution is **paper only, behind six pre-registered go-live gates**; no real order has been placed. Earlier work on the same project covers broadcast-video player tracking.

SELECTED PROJECT — COURTVISION

Calibrated multi-sport forecasting engine, with the validation harness and paper-only execution controls as first-class components.

Evidence: github.com/neelshah/court-vision/blob/master/docs/QUANT_BRIEF.md · github.com/neelshah/court-vision/blob/master/EVIDENCE.md

Pricing quality, scored against the market

- **Pregame vs the Shin-devigged close:** NBA moneyline Brier **0.1735 vs 0.1666** for the close (n=743, 95% CI of the gap includes 0); MLB moneyline 0.2429 vs 0.2390 (n=13,992); soccer O/U 2.5 0.2465 vs 0.2390 (n=7,558); totals and ATP tennis trail by more. **Nothing beats the close.**
- **In-game:** conditioning on realized game state is sharper than a static pregame prior (**NBA Brier 0.209 to 0.159, MLB 0.241 to 0.126**, real-corpus out-of-sample). Where a live market exists (NBA end of Q1, Kalshi MLB and soccer ticks), the market is sharper than the model, and the verdicts are published.
- **Signals:** 0 of 60 candidate signal classes shipped; each was scored with a game-clustered Diebold-Mariano test and Benjamini-Hochberg control, and rejected.

Validation, and what did not survive it

- **Harness:** expanding-window walk-forward with a per-fold date assertion, **truncation-invariance** leak tests, acceptance only when a change beats baseline on **two independent corpora**, and a published minimum detectable effect for every market-relative row.
- **Retractions:** my own harness traced six headline figures to a leak, a grading artifact, an overfit or regime dependence, including an early "+18%" ROI figure (a market-follow artifact, retracted) and a Q4 leak in an in-play metric. The retraction record is public.
- **Execution controls, paper only:** an order lifecycle whose live path is unreachable behind three independent refusals; a venue fee model that fails closed; a **CLV ledger with an off-market-price filter**, with its writer tested race-safe across two OS processes. The forward paper series has not started (0 of 8 required weeks).

Engineering & Data

- **Private engine:** FastAPI with **~99 endpoints across 12 routers, 9 long-running daemons under a watchdog**, a 430-module codebase, **~7,400-test** pytest suite, CI/CD, 5 Docker images; an **MCP server** (interface documented publicly) answers queries from Claude with cited artifacts.
- **Modeling & data:** per-stat **XGBoost** with **isotonic** calibration; four de-vig methods from scratch incl. **Shin (1992)**; a **151-file intelligence layer** incl. a 291,625-pair player-vs-player matchup matrix; Claude agents propose signals and the gate rejects them.
- **Computer vision:** broadcast video to court coordinates on one consumer GPU: YOLOv8 detection → SIFT homography → Kalman+Hungarian tracking → OSNet re-ID reimplemented in PyTorch; output quality measured and documented.
- **Single-concept public repos:** shin-devig, calibration-gate, walkforward-guard, prereg-cards, sdv-calibration-audit (github.com/neelshah)

PROFESSIONAL EXPERIENCE

Data Analyst Intern — Sunsolor · Tempe, AZ · 2025

Built live Tableau / Power BI suites; cut manual reporting latency ~30%; automated daily market-trend pipelines in Python; data-quality audits across 5+ datasets; optimized SQL on 1M+-row datasets.

Data Annotation Specialist — DataAnnotation.Tech · Remote · 2024–2025

Labeled 10,000+ structured samples for production ML training & LLM fine-tuning; authored edge-case documentation cutting pipeline error variance ~20%; refined labeling schemas with data scientists.

Business Intelligence Analyst — Fortrex Securities · Madison, WI · 2023–2024

Built secure executive reporting interfaces (99.9% uptime); refined backend payment APIs & reporting pipelines for high-throughput billing; engineered real-time anomaly-detection alerting.

EDUCATION & CERTIFICATIONS

B.S. Data Science — University of Iowa · 2022–2027, expected May 2027 (3 years coursework completed; finishing remaining credits online). Coursework: math stats, probability theory, machine learning, financial analytics.

Certifications: Certified Data Scientist (USDSI) · ML Specialist & Data Viz (Kaggle) · CBIAP (IABAC).

SKILLS

Languages: Python (expert), SQL, JavaScript/TypeScript (working)

ML/CV: PyTorch, OpenCV, YOLOv8/Ultralytics, ONNX, TensorRT, NumPy, Pandas, scikit-learn, XGBoost, LightGBM, statsmodels

Backend/Infra: FastAPI, WebSocket/SSE, PostgreSQL, SQLite, Docker, GitHub Actions, GCP, Tableau

Methods: walk-forward / leak-free validation, calibration (isotonic), de-vigging (Shin), fractional-Kelly, Monte-Carlo simulation, shadow-logging, multi-agent orchestration